

A Survey of Artificial Intelligence in Multiple Sclerosis MRI: Datasets, Technical Performance, Evidence Quality, and Clinical Translation

Article type: Survey article / critical narrative review (Springer-style manuscript)

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Running title: Artificial intelligence in multiple sclerosis MRI

Manuscript statistics: 6,721; 3 tables; 4 figures

***Draft status:** AI-assisted Springer-style research draft—not submission-ready. Springer Nature journals use journal-specific Instructions for Authors; the target journal's article type, word limit, reference style, figure specifications and AI policy must be verified before submission.*

Abstract

Magnetic resonance imaging (MRI) is indispensable to multiple sclerosis (MS) diagnosis and monitoring, yet manual interpretation is time-consuming, variable and poorly suited to extracting the full longitudinal and quantitative signal. Artificial intelligence (AI) has therefore been applied across the imaging pathway, from reconstruction and harmonisation to lesion segmentation, biomarker detection, diagnosis and disability prediction. This survey critically synthesises work published from January 2015 to 30 June 2026, with selected earlier benchmarks and consensus statements, and organises the field by datasets, clinical tasks and maturity of evidence rather than by individual algorithms. Public MS MRI resources have enabled reproducible white-matter lesion segmentation, but remain small, brain-centred and incompletely representative of routine practice. Large trial and registry archives improve statistical power and scanner diversity while restricting independent reproduction. Across comparable settings, automated brain lesion segmentation often approaches inter-rater agreement, whereas small, cortical, enhancing and spinal lesions remain harder. Central vein sign and paramagnetic rim lesion models are promising, particularly after their incorporation into the revised diagnostic framework, but usually depend on specialised sequences, preselected lesions or limited external testing. Diagnostic and prognostic studies frequently report high discrimination; however, case–control spectrum bias, patient leakage, domain shift, absent calibration and sparse prospective validation substantially reduce clinical credibility. The strongest translation signal is for assistive longitudinal monitoring and quantitative reporting, where retrospective multi-centre and small workflow studies show improved activity detection or shorter reading time. Evidence that AI improves patient outcomes, treatment selection or cost-effectiveness is not yet established. Progress now depends less on marginal benchmark gains than on patient-level data governance, geographically independent validation, calibrated failure-aware systems, prospective human–AI studies and post-deployment monitoring.

Keywords: multiple sclerosis; magnetic resonance imaging; artificial intelligence; datasets; external validation; clinical translation

1. Introduction

Multiple sclerosis is a chronic inflammatory and neurodegenerative disease in which MRI supports diagnosis, assessment of dissemination in space and time, exclusion of mimics, treatment monitoring and characterisation of tissue injury. The revised diagnostic framework expands the interpretive role of imaging by recognising the optic nerve as a fifth anatomical location and by incorporating the central vein sign (CVS) and paramagnetic rim lesions (PRLs) as supportive markers under defined conditions [1]. Its accompanying MRI consensus specifies how these markers and anatomical locations should be acquired and interpreted [2]. Contemporary monitoring guidance also emphasises standardised brain and spinal cord acquisition, consistent positioning, three-dimensional fluid-attenuated inversion recovery (FLAIR) when feasible, and comparable follow-up protocols [3]. These recommendations increase the informational value of MRI but also expose a practical mismatch: clinical decisions increasingly depend on subtle, distributed and longitudinal findings, whereas routine

assessment remains constrained by reader time, variable image quality and imperfect visual comparison.

The attraction of AI is consequently broader than automated lesion outlining. Algorithms may reconstruct undersampled images, flag degraded examinations, register serial scans, synthesise missing contrasts, harmonise scanner effects, quantify tissues and lesions, recognise emerging imaging biomarkers, classify diagnoses and predict future disability. The same breadth creates a risk of conceptual conflation. A model that produces a visually plausible FLAIR image is not necessarily safe for detecting a 3-mm new lesion; a segmentation network that achieves a high Dice score on a hidden challenge set is not thereby a clinically validated monitoring tool; and a classifier that separates established MS from healthy controls does not replicate the diagnostic problem faced in an undifferentiated clinic population.

Previous reviews have often catalogued architectures or pooled performance across heterogeneous datasets. Such summaries can obscure the properties that most strongly determine clinical value: who entered the dataset; how repeated examinations were split; which scanners and sequences were represented; how the reference standard was constructed; whether testing was truly independent; whether probabilities were calibrated; and whether output changed decisions or outcomes. Meta-analytic estimates of diagnostic sensitivity and specificity can appear impressive even when the component studies use artificially clean case–control samples, divergent thresholds and incompletely independent testing [4]. Similarly, systematic reviews of deep learning in MS MRI show rapid technical expansion but recurrent weaknesses in reporting, external validation and clinical anchoring [5].

This survey therefore treats data realism, task definition and evidence maturity as a single evaluative chain. We ask four questions. First, do the available data represent the intended patients, examinations and clinical alternatives? Second, is technical performance assessed with metrics appropriate to lesion size, class imbalance and longitudinal use? Third, does the evidence progress from internal optimisation to independent multi-centre testing and prospective human–AI evaluation? Fourth, is there credible benefit at the level of workflow, clinical decisions, health outcomes or resource use? Our central conclusion is deliberately asymmetric: AI for lesion segmentation, quantitative analysis and longitudinal monitoring is approaching useful assistive deployment, whereas autonomous diagnosis and individualised prognosis remain predominantly at the technical-validation stage.

2. Review methodology

2.1 Search scope and eligibility

We conducted a structured critical review rather than a registered systematic review. Searches covered 1 January 2015 to 30 June 2026 and were supplemented by earlier challenge reports, foundational dataset descriptors, diagnostic criteria and MRI consensus documents. Sources included PubMed/MEDLINE, public IEEE Xplore records, publisher pages, reference and citation tracking, challenge websites, dataset repositories, and public regulatory documents. Search concepts combined terms for multiple sclerosis, MRI, machine learning, deep learning, artificial intelligence,

segmentation, classification, prognosis, longitudinal change, atrophy, spinal cord, cortical lesion, CVS, PRL, harmonisation, validation, workflow and regulation. English-language human MRI studies were prioritised. Studies were included when they described an MS-relevant dataset, developed or evaluated an AI-enabled imaging method, assessed reproducibility or workflow effects, or provided reporting, bias or implementation guidance applicable to medical imaging AI.

We excluded non-imaging AI, purely simulated demonstrations without an MS-relevant downstream evaluation, unverifiable scraped image collections, conference abstracts lacking sufficient methodological detail, and papers whose test data could not be distinguished from training data. Alzheimer disease, ageing, vascular white-matter hyperintensity, stroke and tumour datasets were not treated as MS datasets, even where used for pretraining or control images. Particular attention was paid to mirrors and repackaged challenges so that the same participants were not double-counted. Because database searching and screening were not performed in duplicate and no protocol was registered, the review should not be interpreted as exhaustive or as a formal meta-analysis.

2.2 Evidence classification and appraisal

Evidence was grouped by task and assigned a validation level: subject-level internal cross-validation; held-out testing from the same institution or data ecosystem; leave-site-out testing within a coordinated consortium or trial; geographically and operationally independent external validation; prospective or temporally ordered workflow evaluation; and patient-outcome evaluation. We used CLAIM 2024 to examine imaging-AI reporting [6], TRIPOD+AI and PROBAST+AI to assess prediction-model transparency and risk of bias [7,8], DECIDE-AI for early live clinical evaluation [9], and FUTURE-AI for trustworthy deployment and monitoring [10]. Where randomised trials were discussed, the CONSORT-AI and SPIRIT-AI extensions informed expectations for intervention description, error handling and human interaction [11,12].

No pooled Dice, AUC or accuracy was calculated. These measures are not exchangeable across lesion types, prevalence, annotation policies, MRI protocols, patient spectra or validation levels. Quantitative results are reported only as illustrative ranges or representative findings with their data conditions. We gave greater evidential weight to patient-level splitting, hidden or independent test sets, multi-rater reference standards, external centres, prespecified analyses, calibration, reader studies and clinically consequential endpoints. Commercial involvement was not treated as disqualifying, but was considered alongside data access, comparator choice, adjudication and reproducibility.

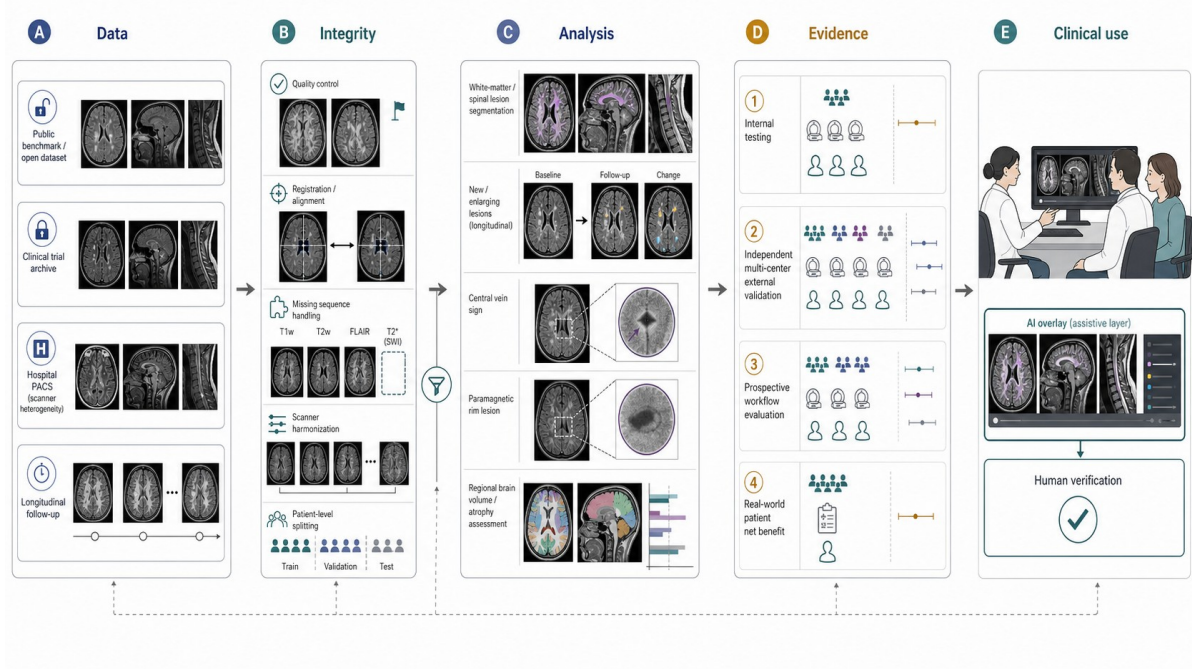


Figure 1. From MS MRI data ecosystems to clinically useful AI. (A) Public benchmarks, controlled clinical-trial archives, heterogeneous hospital PACS examinations and longitudinal follow-up contribute different forms of scale, standardisation and realism. (B) Quality control, registration, missing-sequence handling, scanner harmonisation and patient-level splitting determine whether downstream evaluation is valid. (C) Task-level outputs include brain and spinal lesion segmentation, longitudinal change, CVS, PRL and regional atrophy measurement. (D) Evidence should progress from internal testing to independent multi-centre validation, prospective workflow evaluation and real-world patient net benefit. (E) Deployment is assistive: AI overlays remain subject to clinician verification and shared decision-making. The figure is an original AI-generated scientific illustration created for this survey.

3. Data foundations: datasets, benchmarks and clinical realism

3.1 Public challenges and open resources

Public benchmarks made MS imaging AI measurable, but their contribution is often misread. The historical MICCAI 2008 challenge assembled 45 cross-sectional cases from two 3-T Siemens sites and standardised images through extensive resampling [13]. The ISBI 2015 longitudinal challenge contains 82 sessions but only 19 patients, all from one Johns Hopkins 3-T Philips environment; its 4–6 examinations per patient are valuable for longitudinal methods but create a severe leakage risk if scans rather than patients are randomised [14,15]. MSSEG 2016 expanded acquisition diversity to 53 patients across four scanners at three French sites and obtained seven expert delineations, with a deliberately unseen scanner in testing [16,17]. Its importance lies as much in exposing inter-rater variability as in ranking algorithms.

MSSEG-2 reframed the target from total lesion burden to new lesions between two FLAIR examinations. It includes 100 patient pairs acquired on 15 scanners, with all GE data excluded from training, thereby creating a purposeful compound domain shift [18]. Four neuroradiologists independently marked new lesions before senior consensus. Approximately 40% of new lesions were recognised by no more than two of four experts, demonstrating that small longitudinal targets do not have a noise-free ground truth. The benchmark is clinically closer to follow-up interpretation than cross-sectional segmentation, but remains FLAIR-only and does not encode enlargement, disappearance or treatment consequences.

Other public resources broaden geography and acquisition, although none resolves representation alone. The 3D-MR-MS set provides 30 single-site cases with T1, T2, FLAIR and post-contrast T1 plus jointly revised consensus masks [19]. Long-MR-MS adds 20 patients with one baseline–follow-up interval and labels new, disappearing, enlarging and shrinking lesions, but uses older 1.5-T, 3-mm two-dimensional imaging [20]. MSLesSeg contains 115 examinations from 75 patients across several 1.5-T systems and uses a patient-level challenge split [21]. MS-Baghdad contributes 60 routine 1.5-T examinations from 20 centres together with EDSS and lesion masks, providing valuable heterogeneity but limited protocol metadata and a mildly disabled case mix [22].

Recent datasets illustrate both progress and continuing gaps. SibBMS includes MS or radiologically isolated syndrome cases and healthy controls, but its accessible description is internally inconsistent about how many cases have full lesion masks and about licensing; it should not be assumed that every scan is equivalently labelled [23]. MS3SEG supplies 100 routine 1.5-T cases with a mask that distinguishes MS lesions from other white-matter hyperintensities, but is single-site, excludes lesions below 3 mm and lacks longitudinal outcomes [24]. PediMS provides longitudinal paediatric MRI and clinical metadata, yet only nine patients; its best use is as a difficult out-of-distribution probe rather than as an independent training corpus [25]. Shifts 2.0 repackages ISBI, MSSEG, Ljubljana and private Lausanne data into a common uncertainty benchmark [26]. It is useful for testing domain shift, but cannot be counted as a new cohort and its out-of-domain contrast simultaneously changes centre, scanner, population, lesion burden and annotation procedure.

The public landscape is therefore lopsided. Brain FLAIR white-matter lesion masks are far more available than expert labels for enhancing lesions, cortical lesions, CVS, PRL, atrophy, treatment response, cognition or disability progression. Dedicated spinal resources are particularly limited. A large 30-site study of 1,042 participants demonstrated robust automated cord segmentation but achieved substantially lower agreement for intramedullary MS lesions; the trained system is available through the Spinal Cord Toolbox, whereas the underlying multi-centre images are not a reusable public benchmark [27]. The controlled MS-Multi-Spine initiative is a welcome response, but public protocol records do not yet offer the documentation and open images available for established brain challenges. Dataset availability has consequently shaped the apparent maturity of tasks: the best benchmarked problem is not necessarily the most clinically important problem.

Table 1. MS MRI datasets and benchmark characteristics

Resource	People / sessions	Primary task and labels	MRI	Centres / scanners	Longitudinal	Annotation	Access	Principal limitation
ISBI 2015	19 / 82	WM lesion segmentation; hidden test masks	T1, PD/T2, FLAIR	1 site; 3 T Philips	4–6 visits	2 raters per session	Registered public challenge	Scan count obscures n=19; single scanner; high time-point leakage risk
MSSEG 2016	53 / 53	Cross-sectional WM lesions	3D FLAIR, T1±Gd, PD/T2	3 sites; 4 scanners; 1.5/3 T	No	7 experts; LOP-STAPLE	Controlled request; hidden test	Only 15 labelled training cases; small unseen-scanner subset
MSSEG-2	100 / 200	New-lesion segmentation between paired FLAIR	3D FLAIR	15 scanners; 1.5/3 T; 3 vendors	2 visits, 1–3 y	4 readers plus senior consensus	Controlled challenge; hidden test	FLAIR-only; vendor/site/label shifts entangled; 40% low-agreement lesions
3D-MR-MS	30 / 30	Cross-sectional lesion masks	T1, T2, FLAIR, post-Gd T1	1 site; 3 T Siemens	No	3 experts; joint revision	Public	Small, single scanner; not all source contrasts native isotropic
Long-MR-MS	20 / 40	New, disappearing, enlarging, shrinking lesions	2D T1, T2, FLAIR	1 site; 1.5 T Philips	2 visits	2 experts; consensus	Public	One interval per patient; 3-mm slices; labels depend on registration
MSLesSeg	75 / 115	WM lesion segmentation	T1, T2, FLAIR	Several hospitals; ~3 1.5 T systems	1–4 visits	Two-reader workflow plus senior validation	CC BY 4.0	Most participants and all test cases have one time point
MS-Baghdad	60 / 60	Lesion masks plus EDSS	Routine T1, T2, FLAIR	20 centres; 1.5 T	No	3 radiology/neurology experts	Public	Thick-slice heterogeneity; limited protocol metadata; mildly disabled sample
SibBMS	193 / 193	MS/RIS and healthy controls; partial lesion labels	T1, T2, FLAIR; post-Gd T1 in patients	Single region; scanner incompletely reported	No	Primary description inconsistent about mask coverage	Downloadable; licence wording unclear	MS/RIS pooled; control-source shortcuts; annotation and licence ambiguity
MS3SEG	100 / 100	MS lesions vs other WMH	2D T1, T2, axial/sagittal FLAIR	1 site; 1.5 T Toshiba/Canon	No	Junior–senior consensus; ≥3-mm lesions	CC BY 4.0	Single scanner; excludes sub-3-mm lesions; no outcomes
PediMS	9 / 28	Paediatric lesions and clinical follow-up	T1, T2, FLAIR	One dominant 3 T Siemens site	1–6 visits	Junior rater plus specialist validation	CC BY	Extremely small; irregular intervals; best suited to OOD testing
Shifts 2.0	172 scan samples	Uncertainty distribution shift under	Common T1/FLAIR preprocessed	Repackages public sets plus private Lausanne	Mixed	Source-dataset references	Public parts; private evaluation server	Not an independent cohort; compound population/scanner/label shift
MS PATHS	16,568 enrolled; 14,414 MRI studies	Standardised quantitative MRI plus clinical outcomes	3D MPRAGE + SPACE-FLAIR	10 institutions; Siemens 3 T	3,822 with repeat MRI	No dense voxel masks at scale	Controlled collaboration	Vendor-specific standardisation; care-driven follow-up; industry involvement
MindGlide trial ecosystem	Train 2,871 / 4,247; external 1,001 / 14,952	Tissue/lesion segmentation; longitudinal trial analyses	Mixed legacy clinical/trial contrasts	592 training scanners; multiple trials/cohorts	Extensive	Trial/cohort-derived labels	Mostly proprietary	Large and heterogeneous, but composition and labels limit independent reproduction

Note: People and imaging sessions are separated because repeat examinations are not independent. WM, white matter; Gd, gadolinium; RIS, radiologically isolated syndrome; OOD, out-of-distribution. Counts are tied to the cited descriptors and should not be summed across repackaged datasets.

3.2 Controlled cohorts, clinical trials and routine-care archives

Request-access cohorts provide a different type of realism. The Swiss Multiple Sclerosis Cohort established prospective clinical and radiological follow-up across seven centres with central quality control, although imaging can remain care-driven and protocol substitutions occur [28]. OFSEP links nationwide French clinical, biological and MRI data through a network of specialist centres and Shanoir infrastructure, offering scale and routine variation but also missingness, protocol drift and centre- or treatment-dependent selection [29]. MS PATHS combines standardised 3-T MRI with patient-reported and performance outcomes across ten institutions; by its initial report it had more than 16,000 enrollees and over 14,000 MRI studies, yet its streamlined acquisition was Siemens-specific and its scale does not imply dense manual voxel annotation [30]. The NAIMS 7-T repository pools 528 scans from 350 people across five institutions, enabling study of cortical and chronic active lesions but reflecting a selected research population and specialised high-field protocols [31].

Clinical-trial archives can contain thousands of serial scans, rigorous visit schedules and treatment assignments. MindGlide, for example, was trained on 4,247 scans from 2,871 patients acquired on 592 scanners across trials and a cohort, and evaluated on 14,952 scans from 1,001 people in independent trials and paediatric routine care [32]. Such scale supports rare acquisition conditions, longitudinal consistency and recovery of group-level treatment effects. It does not automatically permit independent replication: trial composition, labels and preprocessing are often proprietary, and participants are selected by eligibility criteria that narrow comorbidity, disease stage and adherence. A model can therefore be internally robust within the clinical-trial ecosystem while remaining uncertain in emergency, community, low-resource or highly comorbid populations.

Routine PACS archives reverse this trade-off. They contain motion, incomplete protocols, thick slices, scanner upgrades, unconventional positioning and undocumented clinical decisions—precisely the variation a deployed tool will encounter. They also contain informative missingness and correlated care processes: the sequence ordered, contrast administered and frequency of follow-up may reflect a clinician's prior belief about activity. Without careful design, an algorithm can exploit these process signatures instead of disease biology. De-identification can remove dates needed for longitudinal reasoning; repeated examinations can leak across partitions; and reports used as labels can reproduce reader error. A realistic dataset is therefore not simply a large dataset. It is one whose sampling, missingness, labels and intended use are explicit enough to support a defensible target population and evaluation.

Figure 2 locates these resources along dimensions that are often collapsed into a single sample-size claim. Open benchmarks maximise comparability but usually occupy a narrow acquisition domain; controlled cohorts add scale and longitudinal depth; routine archives add operational realism at the cost of incomplete metadata and noisier reference standards. The resulting design problem is not to select one “best” source, but to combine complementary sources while preserving patient-level independence and documenting anatomy, temporal depth, sequences and annotation.

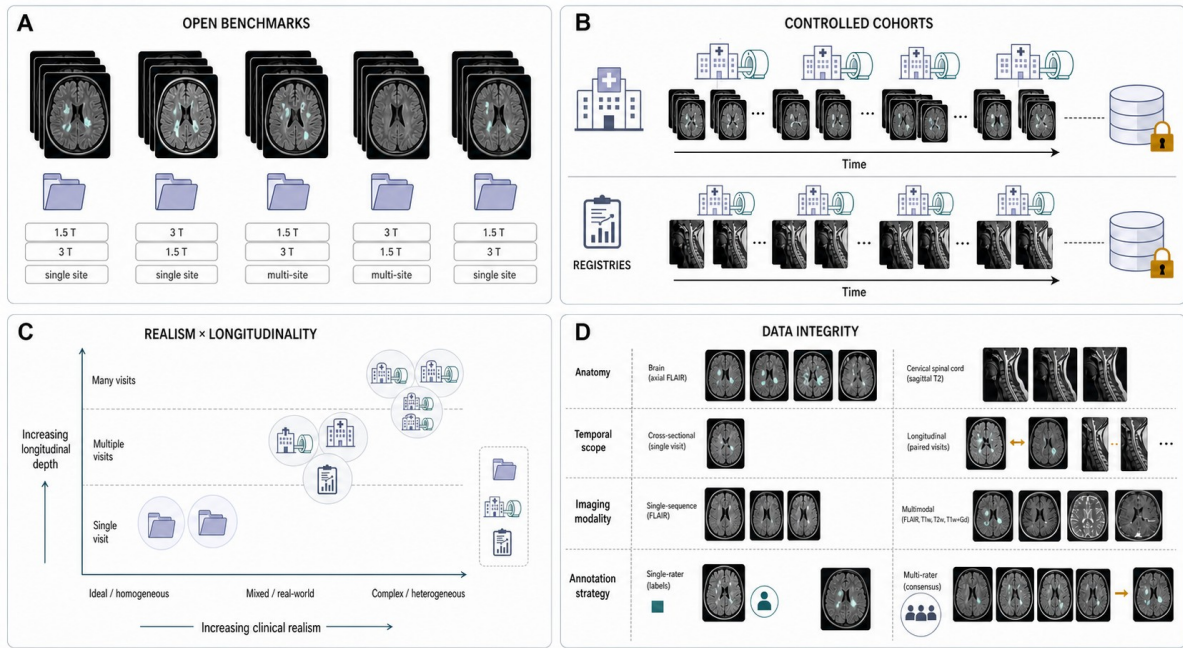


Figure 2. Dataset scale, access and realism are complementary rather than interchangeable. (A) Open benchmarks support transparent comparison but commonly contain relatively small, highly curated cohorts. (B) Controlled clinical-trial and registry collections add multicentre and longitudinal depth while limiting unrestricted reproduction. (C) Clinical realism and longitudinal depth vary independently, so sample size alone does not define fitness for deployment. (D) Data integrity depends on anatomy, temporal scope, sequence coverage and annotation strategy; repeated visits must remain grouped by patient. The figure is an original AI-generated scientific illustration created for this survey and is conceptual rather than a quantitative map of named datasets.

3.3 Annotation, leakage and hidden shortcuts

Expert annotation should be treated as a measurement process, not as ground truth delivered without error. Lesion boundaries vary with windowing, slice thickness, registration and rules for punctate or confluent abnormalities. Consensus methods can improve stability but also hide disagreement. In MSSEG, performance depends on which expert or fused reference is used; in MSSEG-2, disagreement is concentrated in the small lesions that matter for activity classification. Reported Dice should therefore be accompanied by lesion-wise sensitivity and precision, false-positive lesions per scan, volume bias, size-stratified results and performance in cases with no activity. For longitudinal tools, a patient-level category such as no, few or several new lesions can sometimes align better with the clinical decision than subvoxel overlap.

Patient-level splitting is non-negotiable. When serial scans of the same individual enter training and testing, a model can recognise anatomy, lesion topology, acquisition settings and preprocessing artefacts. The same principle applies to patch extraction: random patches from one volume are not independent observations. Centre-level separation is stronger, but “leave-one-centre-out” within a centrally coordinated trial still shares eligibility, protocol, preprocessing and label governance. We therefore distinguish it from geographically and operationally independent testing. Scanner diversity likewise should not be reduced to a count; vendor, field strength, coil, reconstruction software, sequence parameters and patient population need to be disentangled or at least documented.

Diagnostic datasets are especially vulnerable to shortcuts. Healthy controls are frequently obtained from separate research programmes or sites. A classifier may then identify scanner, age or acquisition

differences rather than MS. OASIS, ADNI, IXI, BrainWeb, BraTS, ISLES and the WMH Challenge may support pretraining or methodological experiments but are not MS cohorts. Unverified Kaggle images and two-dimensional exports can obscure patient identity, duplicate cases, remove metadata and sever images from expert ascertainment. Their inclusion should trigger provenance analysis rather than be interpreted as inexpensive external validation.

4. Artificial intelligence methods and applications across MS MRI

4.1 Reconstruction, quality control, registration and harmonisation

AI acts first on the measurement chain. Motion detection, artefact classification, automated positioning and protocol-completeness checks can prevent unusable data from entering downstream models. Registration of serial scans supports subtraction imaging and consistent regional measurement, but registration error around ventricles, atrophy and resection-like tissue changes can mimic lesion change. A quality-control model should therefore issue failure states and visual overlays rather than silently normalise every input. Missing-sequence synthesis is attractive where post-contrast, susceptibility-sensitive or high-resolution images are unavailable, yet a synthetic sequence is an estimate conditioned on observed data. It cannot be assumed to contain a lesion or vein that was not identifiable in the source contrasts.

DeepHarmony learned contrast transformation from overlapping subjects to reduce scanner-related inconsistency [33], while HACA3 generalised harmonisation across 21 sites, field strengths, vendors and flexible contrast combinations and evaluated downstream lesion and volumetric measures [34]. These studies illustrate an appropriate validation principle: image similarity is insufficient; harmonisation should preserve known pathology, reduce unwanted scanner variance and maintain longitudinal biological change. Super-resolution work on 429 images across six datasets improved PSNR, structural similarity and subsequent lesion segmentation after simulated degradation [35]. Nevertheless, simulated low resolution is easier to characterise than real motion, noise and interpolation. Generative reconstruction that improves appearance can smooth a tiny lesion or invent high-frequency structure. Travel-subject studies, test–retest measurement and lesion-level non-inferiority are more clinically relevant than visual preference alone.

4.2 White-matter, enhancing and spinal lesion segmentation

White-matter lesion segmentation is the most technically mature task. In approximately 1,000 phase III trial scans, a fully convolutional approach reported high tissue overlap and a T2-lesion Dice around 0.82 under leave-centre-out evaluation [36]. The result demonstrates scalable measurement under coordinated acquisition, not unrestricted generalisation. Models trained on smaller institutional data and tested on MSSEG have at times matched or exceeded the average single rater relative to a fused consensus [37]. The careful interpretation is that algorithm–consensus agreement can fall within the human inter-rater range; it does not show that an algorithm is clinically superior to expert interpretation.

Methods designed for acquisition variability provide a more informative stress test. A contrast-adaptive model produced brain structures and lesions across four disparate datasets without protocol-

specific retraining [38]. BIANCA-MS reported an aggregate Dice around 0.72 but with a standard deviation of 0.25 and substantial false-positive and false-negative cluster counts [39]. In a small real-world external evaluation, median Dice fell from 0.73 internally to 0.66 in a Latin American cohort, while human inter-rater Dice was 0.63 and test–retest consistency remained high [40]. The broad lesson is not that 0.66 is universally sufficient. It is that external loss, reference uncertainty and repeatability must be considered jointly, and that lesion burden and clinical tolerance for false positives determine whether a segmentation is useful.

Large private trial data improve robustness but do not erase the domain gap. A multi-arm U-Net trained in multi-centre relapsing MS trials achieved internal Dice at or above 0.66 and sensitivity at or above 0.72; on an external challenge these fell to 0.62 and 0.68, respectively, although group differences and lesion–disability associations were recovered [41]. This is valuable for trial end points and population-level research. It remains different from reliable per-patient detection of a single actionable lesion in an uncurated PACS examination.

Enhancing lesions are comparatively rare and small. In 372 scans from the Swiss cohort, a model achieved Dice 0.76 and a high true-positive rate under patient-level cross-validation, but lacked a fully independent external cohort [42]. Another system developed with 600 examinations and compared lesion counts in 2,846 routine scans reported Dice 0.698, lesion sensitivity 0.844 and a false-discovery rate of 0.307; performance improved markedly for lesions larger than 100 voxels [43]. These size effects matter because an average patient-level Dice can be dominated by larger lesions while the clinically consequential punctate enhancement remains missed or overcalled.

Spinal cord analysis separates easy anatomy from difficult pathology. A 30-site network reached a cord Dice of approximately 0.95, but spinal lesion Dice was about 0.60 despite sensitivity of 0.83 and precision of 0.77 [27]. Partial-volume effects, small cross-sectional area, motion, pulsation, susceptibility and heterogeneous axial or sagittal protocols explain part of the gap. A recent multi-contrast system trained on 4,428 annotated images from 1,849 patients across 23 centres illustrates the value of heterogeneous training and explicit external assessment [44]. Even so, missing contrasts and site-specific annotation policies make spinal lesion segmentation less mature than cord morphometry. A clinically safe tool must distinguish uncertain image quality from confident absence of lesions.

4.3 Cortical lesions, CVS and PRL

Cortical and chronic active lesions are biologically and diagnostically important but impose specialised acquisition and labelling requirements. At 7 T, the CLAIMS network was trained on 2,247 consensus cortical lesions from 60 patients and externally assessed in 20 patients; lesion detection was approximately 74% with a 30% false-positive rate, and intracortical lesions were harder than leukocortical lesions [45]. A larger 3/7-T multi-institution study found lesion F1 around 0.64 in-domain and around 0.50 out-of-distribution, with small lesions and leukocortical boundaries driving error [46]. Increasing model capacity did not reliably solve the domain problem. These results caution against interpreting high-field research performance as readiness for routine 3-T clinical FLAIR.

CVS automation has a direct conceptual link to the revised diagnostic criteria, but most systems remain partly conditioned on manual lesion selection. CVSnet used data from three centres and four

scanners; in an independent test of 314 presegmented lesions, lesion-level AUC was 0.90 and a 50% CVS rule yielded subject-level sensitivity 0.89 and specificity 0.92 [47]. Because non-evaluable lesions were excluded and candidate lesions were manually defined, this is not an end-to-end diagnostic evaluation. A systematic review also shows that CVS performance depends on sequence, field strength, lesion selection and the patient-level rule [48]. The diagnostic validity of CVS should not be conflated with the engineering validity of automated CVS detection.

PRL models face severe class imbalance. RimNet analysed 462 rim-positive and 4,857 rim-negative lesions from 124 patients and reported AUC 0.943, sensitivity 0.706, specificity 0.949 and positive predictive value 0.569 [49]. Expert lesion-level F1 remained substantially higher than the model's, and some small or confluent lesions were excluded. QSMRim-Net achieved sensitivity around 0.68 at specificity 0.99 in 172 patients, but relied on fivefold cross-validation and only 177 positive lesions [50]. High specificity is readily amplified by thousands of negative lesions; positive predictive value, false positives per patient and external-reader comparison are indispensable. An automated multi-biomarker pipeline has reported cross-validated AUROCs around 0.95 for lesions, 0.91 for PRL and 0.87 for CVS in 97 patients, but still awaits independent testing [51].

The clinical premise for these markers is stronger than it was several years ago. Multi-centre work combining cortical lesions, CVS and PRL has shown high specificity for MS and an association between multiple PRLs and progression independent of relapse activity, although the latter estimate was imprecise [52]. The revised criteria create a use case for reliable automated counts and proportions, not a licence for autonomous diagnosis. Prospective studies must test ordinary 3-T sequences, real mimics, non-evaluable lesions and the complete pipeline from image quality through lesion detection to patient-level interpretation.

Figure 3 places these targets in their native imaging context. The visual hierarchy is clinically important: global lesion burden and longitudinal change are not interchangeable with enhancement, cortical pathology, susceptibility markers or spinal cord lesions. Each target requires different sequences, annotation rules, error tolerances and units of analysis, so a single aggregate “segmentation performance” statement cannot describe the field.

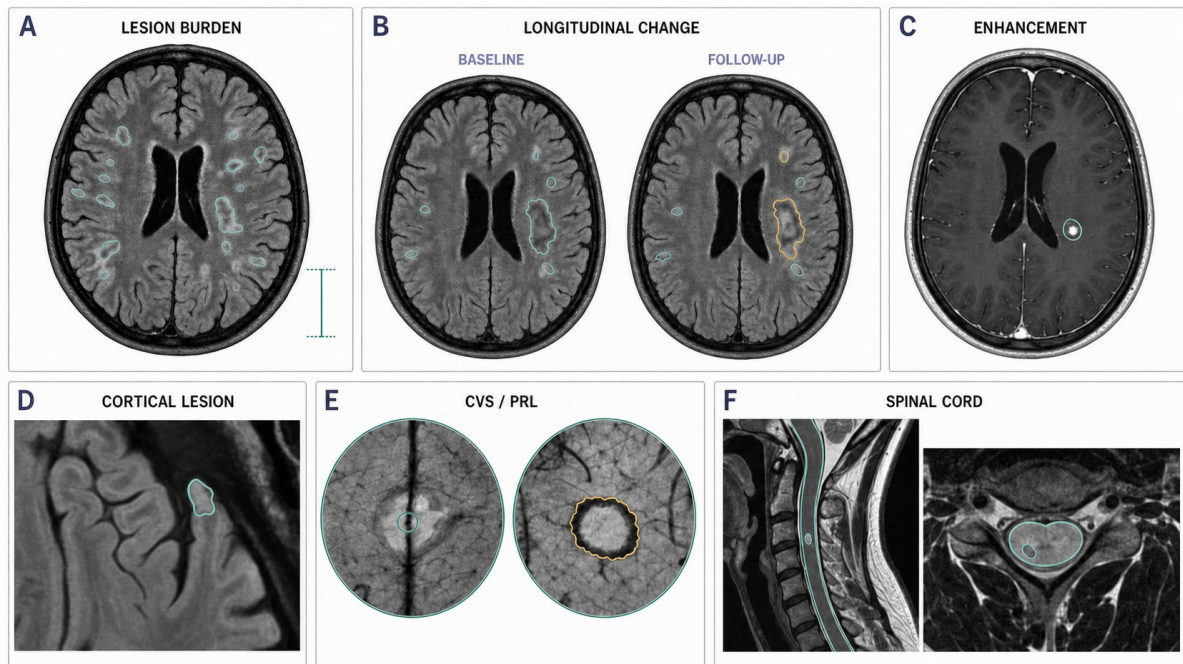


Figure 3. MRI targets for artificial intelligence in multiple sclerosis. (A) Cross-sectional white-matter lesion burden with editable segmentation contours. (B) Paired baseline and follow-up FLAIR images showing new and enlarging lesions. (C) A contrast-enhancing lesion on post-contrast T1-weighted imaging. (D) A cortical lesion at the grey–white matter interface. (E) Susceptibility-sensitive examples of the central vein sign and a paramagnetic rim lesion. (F) Sagittal and axial cervical cord views with cord and intramedullary lesion contours. Overlays are schematic and do not represent patient data or measured performance. The figure is an original AI-generated scientific illustration created for this survey.

4.4 Diagnosis and differential diagnosis

Diagnostic modelling illustrates how the data-generating process can dominate performance. A CNN distinguishing 76 established MS cases from 71 healthy controls reached 87% accuracy [53]. Such a design asks whether two curated groups differ, not whether a new patient meets diagnostic criteria. A more relevant study included migraine, MS, neuromyelitis optica spectrum disorder (NMOSD) and central nervous system vasculitis, reporting one-versus-rest accuracy of 98.8% for MS in a 90-person held-out set [54]. However, the test set came from the same data ecosystem, class proportions were artificial, and radiologists were restricted to selected MRI information while clinical diagnosis normally integrates symptoms, cerebrospinal fluid and serology.

Direct MS–NMOSD discrimination remains challenging. A single-centre two-dimensional FLAIR network in 86 MS and 70 NMOSD cases achieved accuracy 0.761 and AUC 0.85 without external validation [55]. Texture and radiomics models can identify distributed abnormalities, but small feature-to-sample ratios and unstable preprocessing threaten reproducibility. For example, an early radiomics study trained on 36 MS cases and 36 controls showed an external T1-based AUC around 0.78 but sensitivity only 0.30 [56]. High accuracy in balanced data is therefore less informative than sensitivity, specificity, predictive values and calibration at the intended clinic prevalence.

Recent multi-biomarker models based on manually assessed CVS, cortical lesions and PRLs offer a better biological target and have achieved high balanced accuracy in small external samples [57]. Yet they test the combination of expert-read biomarkers, not a complete AI system. A clinically valid diagnostic tool must begin with consecutive or otherwise representative suspected cases, predefine how indeterminate scans are handled, compare against an appropriate multidisciplinary reference

standard, and report subgroup performance across age, sex, race and ethnicity, comorbidity, disease duration, scanner and referring context. It should also be evaluated as an adjunct: the relevant question is usually whether calibrated AI assistance reduces error or unnecessary testing, not whether an image-only model can outperform a clinician deprived of clinical data.

4.5 Longitudinal activity, treatment response and quantitative monitoring

Longitudinal comparison is where automation most directly addresses a routine bottleneck. Registration, subtraction and lesion matching can make new or enlarging lesions more visible and produce structured counts and volumes. A 60-patient study reported new-lesion Dice around 0.83 and lesion sensitivity around 83%, but was single-centre and included relatively few active cases [58]. On the heterogeneous hidden MSSEG-2 test set, a lesion-aware augmentation method reached Dice 0.510 and lesion F1 0.552 in cases with new lesions while keeping false positives low in stable cases [59]. The contrast with easier cross-sectional segmentation shows why activity should be evaluated at lesion and patient levels, particularly in scans with zero events.

The definitive MSSEG-2 analysis is especially instructive: the best method and best expert both achieved approximately 85% accuracy when patients were classified as having no, one to two, or more than two new lesions, even though lesion-level overlap and expert agreement varied widely [18]. This does not mean the algorithm replaces the reader. It suggests that clinically aligned categories can be robust despite uncertain boundaries, provided the system displays candidates and uncertainty for human verification.

Real-world retrospective evidence is emerging. In 397 multi-centre scan pairs, an AI system detected new or enlarging FLAIR activity with patient-level sensitivity 93.3%, compared with 58.3% in routine reports and 85.0% in a core laboratory, with similar high specificity [60]. This is one of the strongest translation signals in the field, but it was a retrospective discrepancy-adjudication study with commercial conflicts and no patient-outcome endpoint. A PACS-compatible evaluation in 54 scan pairs and three experts primarily established usability and reader acceptance [61]. A four-reader crossover study reported 31.5% shorter baseline and 38.4% shorter follow-up reading times, yet one reader did not benefit, another became slower on baseline scans, and software failed in 10% of follow-up cases [62]. In 35 consecutive patients, AI assistance shortened one resident's mean interpretation time by 2.83 minutes, but fixed reading order, a single reader and the small sample allow learning and recall effects [63]. Another small study found additional true new lesions only after readers rejected 61 false-positive AI candidates, showing that sensitivity gains can be exchanged for verification burden [64]. A later retrospective re-read of 83 patients reported more newly identified and enlarging lesions and documented treatment adjustments in 10.8%; a decision change is not proof that the change was appropriate or beneficial [65].

Brain tissue and lesion volumetry can complement activity detection, particularly when visual assessment of slow atrophy is insensitive. Large automated pipelines can produce parenchymal fraction, cortical and deep grey-matter volumes, lesion load and spinal cord area. Scanner and protocol effects can be similar in magnitude to annual biological change, so normative z scores and longitudinal rates require stable acquisition, software version control and uncertainty limits.

DeepBVC learned brain volume change from SIENA-derived pseudo-labels and showed technical agreement, but did not demonstrate a clear association with EDSS progression in its test data [66]. Quantitative output is therefore best treated as a measurement with provenance and confidence—not as a self-interpreting disease score.

Table 2. Task-specific technical performance and evidence limitations

Task	Clinically relevant output	Appropriate metrics	Illustrative performance under stated conditions	Highest common validation level	Recurrent failure mode
Reconstruction / harmonisation	Pathology-preserving image or stable measurement	Downstream lesion/volume error; test-retest; site variance	Super-resolution in simulated low-resolution FLAIR: PSNR 22.7 → 24.1 dB; SSIM 0.64 → 0.72	Multi-dataset technical validation	Hallucinated or smoothed small lesions; paired-data dependence; protocol drift
Brain WM lesion segmentation	Lesion map, count and volume	Dice plus lesion sensitivity, PPV, FP/scan, volume bias	Same-domain Dice commonly ~0.70–0.80; one real-world study 0.73 internal vs 0.66 external; inter-rater 0.63	Small independent external sets	Small/infratentorial lesions, low burden, vascular WMH, unfamiliar scanners
New/enlarging lesions	Patient activity and candidate changes	Lesion F1; FP in stable cases; patient activity category	MSSEG-2 best method and expert each ~85% for 0, 1–2, >2 new lesions; lesion overlap much more variable	Hidden multi-scanner challenge; retrospective PACS	Registration error, tiny lesions, protocol mismatch, false-positive review burden
Gd-enhancing lesions	Active enhancing lesion candidates	Lesion sensitivity, FDR, FP/scan, size strata	Representative Dice ~0.70–0.76; lesions >100 voxels materially easier	Patient-level internal validation; routine retrospective comparison	Rare positives, vessels, motion, tiny punctate enhancement
Spinal cord and lesions	Cord area and intramedullary lesion map	Cord/lesion Dice; lesion precision and sensitivity; QC failure	30-site cord Dice 0.95; MS lesion Dice 0.60, sensitivity 0.83, precision 0.77	Multi-site internal/external technical testing	Partial volume, motion/pulsation, small cord, missing axial or STIR/PSIR contrasts
Cortical lesions	Cortical lesion candidates by subtype	Lesion F1/sensitivity, FP rate, subtype performance	7 T detection ~74% with ~30% FP; multi-institution F1 ~0.64 in-domain vs ~0.50 OOD	Very small external high-field cohorts	Intracortical lesions, leukocortical boundary, 7 T/3 T domain shift
CVS	CVS-positive lesion proportion / rule	Lesion AUC plus patient sensitivity, specificity, non-evaluable rate	CVSnet AUC 0.90; 50% rule sensitivity 0.89, specificity 0.92 after manual lesion selection	Small multi-scanner held-out test	Manual candidate dependence; sequence and eligibility filtering
PRL	Rim-positive lesion count	Sensitivity, specificity, PPV, FP/patient; reader agreement	RimNet AUC 0.943, sensitivity 0.706, specificity 0.949, PPV 0.569; cross-validation dominant	Internal cross-validation / limited multi-centre	Extreme class imbalance, confluent/small lesions, specialised susceptibility imaging
Diagnosis / differential	Calibrated probability within a suspected-case spectrum	Sensitivity, specificity, PPV/NPV, calibration, decision curve	MS vs healthy controls may exceed 0.85 accuracy; MS vs NMOSD example AUC 0.85; pooled estimates are spectrum-sensitive	Mostly same-ecosystem hold-out; few small external sets	Healthy-control shortcuts, artificial prevalence, restricted mimics, missing clinical/CSF context
Atrophy / tissue quantification	Longitudinal volume change with uncertainty	Bias/limits of agreement, test-retest, annualised change, QC rate	High cross-sectional tissue Dice and correlations are common; annual change remains scanner/protocol sensitive	Large trial validation; limited routine prospective use	Scanner upgrade, segmentation drift, pseudo-label bias, threshold disagreement
Disability / PIRA / cognition	Calibrated future risk at a prespecified horizon	AUC plus calibration, Brier score, RMSE, net benefit	Systematic review: median AUC 0.78; pooled EDSS RMSE 1.31 with I ² =95%	Only a minority independently external	Unstable outcome definitions, treatment confounding, small events, absent calibration

Note: Values are representative findings under specific datasets and validation levels, not pooled performance ranges. Dice and AUC must not be compared across tasks, lesion prevalences, label policies or test populations. PPV, positive predictive value; FP, false positive; FDR, false-discovery rate; CVS, central vein sign; PRL, paramagnetic rim lesion; PIRA, progression independent of relapse activity.

4.6 Phenotyping, disability, PIRA and cognition

Prediction is often presented as the ultimate goal of imaging AI, but the evidence is less mature than for measurement. A 2026 systematic review and meta-analysis of 21 studies and 12,252 patients found median classification AUC 0.78 (range 0.57–0.86), median EDSS regression RMSE 1.08 and pooled RMSE 1.31 with 95% heterogeneity; only six studies used external validation, and calibration, decision curves and code were uncommon [67]. Deep learning did not show a stable advantage over conventional machine learning. An EDSS error of more than one point can cross clinically important thresholds, and discrimination alone does not indicate whether a predicted probability is trustworthy.

Individual studies illustrate why attractive headline results need context. In 465 placebo-arm trial participants, raw multimodal MRI predicted one-year confirmed disability progression with AUC about 0.66, rising to about 0.70 after lesion masks were supplied [68]. A multi-centre study reported 83.3% accuracy for two-year EDSS worsening and 85.7% for a composite endpoint, but the held-out test comprised only 48 participants from the same network and calibration was not reported [69]. A radiomics model associated MRI features with disability in an external centre with a correlation around 0.73, yet the endpoint was cross-sectional EDSS rather than future decline [70]. Cross-sectional disability recognition and longitudinal prognosis should not be conflated.

The same distinction applies to cognition. In 540 multi-centre patients, XGBoost classified Symbol Digit Modalities Test impairment with AUC around 0.74, with thalamic, cortical, hippocampal and lesion volumes contributing [71]. This is evidence of contemporaneous phenotype recognition, not prediction of future cognitive deterioration. For PIRA, the target itself is unstable. A systematic review found reported PIRA proportions ranging from 4% to 24% under different definitions [72], and a large MSBase comparison of hundreds of operational definitions showed substantial variation in event rate and persistence [73]. A model can be statistically precise yet clinically incoherent if its label changes with visit schedule, relapse window, EDSS confirmation rule or missing assessments.

A three-centre study of 719 newly diagnosed patients used the first two years of data to predict PIRA in the third year and achieved a test AUC around 0.75; 24-month EDSS and age at onset were more influential than imaging [74]. Multimodal work incorporating MRI, neurofilament light and glial fibrillary acidic protein found spinal cord atrophy to be a stable predictor across two independent cohorts, but remained a high-dimensional feature-discovery analysis rather than a calibrated clinical tool [75]. Future work must separate prognosis under observed treatment from treatment-effect prediction. The latter requires a counterfactual design and cannot be inferred from ordinary risk association.

Unsupervised phenotyping provides a stronger example of scale and independent replication. SuStaIn identified cortex-led, normal-appearing-white-matter-led and lesion-led MRI patterns in 6,322 patients and replicated them in 3,068 independent patients; the lesion-led pattern was associated with greater relapse and progression risk [76]. Treatment-response differences were exploratory subgroup findings rather than prescriptive evidence. A later latent disease-state model used approximately 8,000 trial participants, 118,000 visits and more than 35,000 MRI examinations and validated prespecified properties in independent trials and a real-world cohort [77]. These studies may refine disease

taxonomy, but a reproducible state label is not equivalent to an individually calibrated prognosis or a treatment recommendation.

5. Technical performance, evidence quality and reproducibility

5.1 Validation hierarchy and metric fitness

Validation labels should describe operational independence, not merely a random split. Subject-level cross-validation is appropriate for early development if preprocessing, augmentation and hyperparameter selection are nested. A same-institution hold-out estimates performance on the local pipeline. Leave-site-out testing within a coordinated trial adds scanner variation, but may still share eligibility, annotation and central processing. Geographically independent external validation changes institutional practice and is more persuasive. Temporal validation tests protocol drift. Prospective reader studies reveal interaction effects and automation bias. Only trials or robust quasi-experiments with patient-relevant endpoints can establish clinical benefit.

Metrics should follow the intended decision. Dice rewards spatial overlap and is sensitive to lesion volume; it can be unstable for tiny lesions and undefined in negative cases. Lesion-wise sensitivity should be paired with false-positive lesions per scan and positive predictive value. Volume agreement needs bias and limits of agreement, not correlation alone. Diagnostic and prognostic models require sensitivity, specificity, predictive values at relevant prevalence, calibration intercept and slope, Brier score and decision-curve analysis. AUC is threshold-independent discrimination, not an estimate of clinical utility. Confidence intervals should be clustered by patient and, where relevant, site. Subgroups and failure cases should be prespecified rather than selected after inspection.

5.2 Bias, explainability and reproducibility

Patient leakage, spectrum bias, class imbalance and treatment-related confounding recur throughout the literature. Preprocessing can leak test-set statistics through intensity normalisation, template construction or site harmonisation. Extensive augmentation cannot create unobserved clinical diversity. Labels derived from reports or legacy software transfer their errors, while pseudo-labels can constrain a new model to imitate the old method. Missing data may be informative. Code release helps, but reproducibility also requires trained weights, preprocessing specifications, versioned dependencies, inference hardware, thresholds and test data access. Where data cannot be shared, executable containers, federated evaluation and locked challenge servers can reduce but not eliminate the gap.

Explainability does not repair an invalid design. Saliency maps can reveal that a model attends to ventricles, lesions or image borders, but they are unstable and do not establish causal reasoning. More useful safety mechanisms include input-quality checks, out-of-distribution detection, calibrated uncertainty, lesion-level candidate display, comparison with prior examinations and explicit abstention. The reader must be able to distinguish “no lesion detected” from “image or model not reliable”. Reporting should document who can override output, how failures are logged, and which cases require manual review.

Commercial studies often have access to larger and more realistic archives and can therefore produce valuable translational evidence. At the same time, proprietary labels, undisclosed exclusions, sponsor involvement in adjudication and lack of independent replication can increase risk of bias. Conflict of interest should prompt scrutiny of design and transparency rather than automatic dismissal. Independent post-market evaluation is especially important when software is updated, retrained or deployed across scanners not represented in regulatory testing.

Figure 4 summarises the evidential discontinuities that headline metrics conceal. Validation scope must expand as acquisition and patient heterogeneity increase; generalisation and uncertainty should be assessed together; overlap metrics must be supplemented by calibration and decision analysis; and deployment must be evaluated as a monitored human–AI system rather than as a static model endpoint.

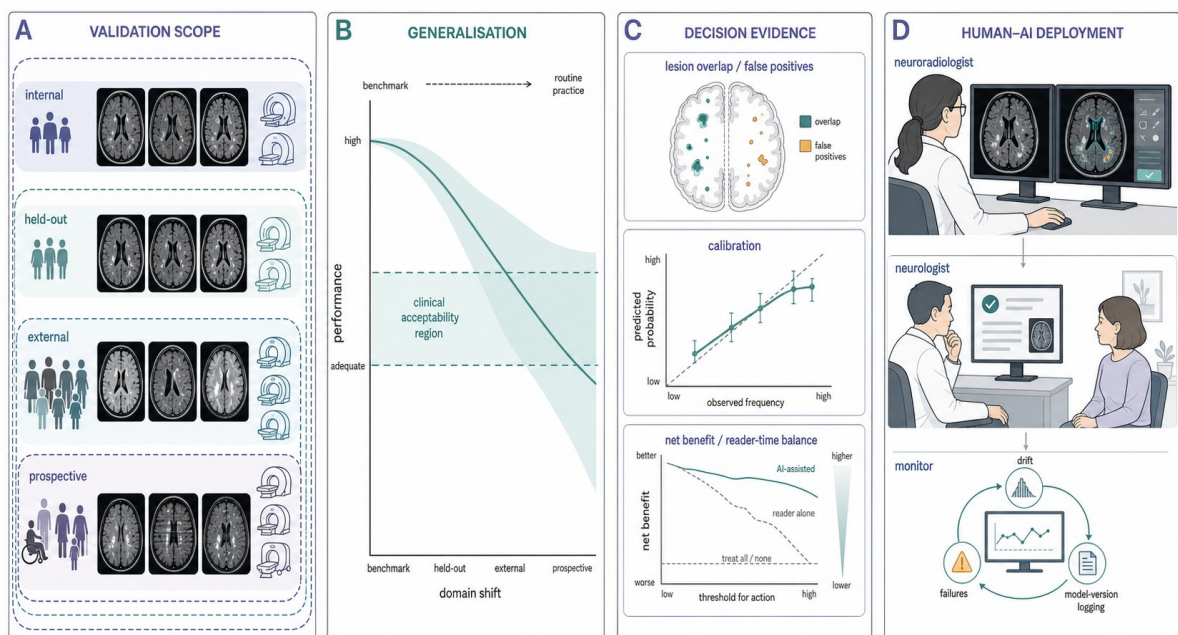


Figure 4. Evidence requirements from technical validation to monitored human–AI deployment. (A) Validation becomes more persuasive as it moves from internal development to held-out, independent external and prospective routine-practice cohorts. (B) Under domain shift, performance may decline while uncertainty widens; acceptability must be defined for the intended use. (C) Spatial overlap and false positives should be supplemented by calibration and clinical decision or workflow analyses. (D) Clinical deployment requires editable output, neuroradiologist verification, multidisciplinary interpretation, patient-centred decisions and post-deployment monitoring of drift, failures and model versions. Curves are conceptual and contain no empirical performance values. The figure is an original AI-generated scientific illustration created for this survey.

6. Clinical translation and deployment

6.1 Regulatory status and intended use

Technical validation, regulatory authorisation, workflow gain and patient benefit are distinct milestones. Regulatory clearance generally establishes substantial equivalence or specified performance for an intended use; it does not prove improvement in relapse control, disability or quality of life. Public FDA records describe lesion and brain-structure quantification functions for products such as Pixyl.Neuro, icobrain and newer NeuroQuant versions, with defined input sequences and performance testing [78–80]. The cleared indication, software version and compatible acquisition

must be checked locally. For example, the icobrain longitudinal indication specifies matched scanner, protocol and contrast conditions. A model used outside that envelope—for example, a different field strength, paediatric population or synthesised sequence—enters a new risk context.

6.2 Human–AI workflow and PACS integration

The near-term value proposition is assistive. Automated lesion maps can pre-populate structured reports, highlight interval change and reduce repetitive measurement. Brain and spinal cord morphometry may make subtle longitudinal change visible. These functions can reduce reading time or increase sensitivity, but can also increase verification burden if false-positive candidates are numerous. Reader studies should measure total task time, edits, disagreements, confidence, missed activity, overcalls and downstream recommendations. Randomisation or counterbalanced reading order is important because the same reader rapidly learns the case. The comparator should be current best practice, which may include subtraction imaging or prior reports rather than unaided visual reading.

Integration is as important as accuracy. A deployable service must receive the correct study and prior, verify sequence identity, handle incomplete examinations, preserve registration quality, return results to PACS or the reporting system, and expose provenance. Failure notifications should reach a named owner. Human verification should be efficient: editable contours, matched prior and current lesions, and concise uncertainty cues are preferable to opaque scores. Responsibility remains shared among manufacturer, institution and clinician; local governance must specify whether output is preliminary, what must be reviewed, and how discordance is resolved.

6.3 Equity, economics and lifecycle monitoring

Equity and cost are under-evaluated. Multi-centre does not necessarily mean demographically representative, particularly when public datasets originate from European or North American academic centres and trial eligibility excludes complex patients. Performance should be stratified across sex, age, race and ethnicity where legally and scientifically appropriate, disability, lesion burden, paediatric or late-onset disease, comorbidity, geography and scanner access. Cost-effectiveness requires more than minutes saved. It should include integration, computing, maintenance, false-positive work-up, repeat imaging, over-treatment, missed disease activity, cybersecurity and staff training. NICE concluded that the clinical and patient-benefit evidence for quantitative MS MRI software was limited and called for prospective comparative studies [81]. A manufacturer-associated simulation projected quality-adjusted life-year gains and annual savings, but omitted several major treatment, imaging and software costs and relied on strong switching assumptions [82]. Such models are hypothesis-generating, not evidence of realised cost-effectiveness. A system that improves sensitivity at the cost of many non-actionable alerts may have negative net value.

Post-deployment monitoring closes the evidence loop. Institutions need baselines for data quality and performance, logs of failure and override, periodic review against manually adjudicated samples, and triggers for investigation after scanner upgrades or protocol changes. Model updates require version control and assessment for regression, subgroup drift and changed calibration. Federated learning may

broaden representation without centralising data, but heterogeneous labels, privacy leakage, unequal site contribution and validation of each update remain unresolved. The trustworthy endpoint is not a model that never changes; it is a controlled learning system whose changes are visible, tested and reversible.

Table 3. Clinical translation readiness matrix

Application	Technical validation	Independent cross-centre evidence	Prospective / workflow evidence	Regulatory position	Patient net-benefit evidence	Readiness judgement
Brain WM lesion segmentation	High	Moderate; external loss is consistent	Limited reader studies	Several cleared quantitative tools	Absent	Near clinical use as editable quantitative assistance
Longitudinal new/enlarging lesion monitoring	High for candidate detection	Moderate; retrospective multi-centre signal	Small crossover and time studies; failure burden documented	Included in some cleared products under constrained inputs	No completed outcome trial	Strongest translational case as supervised second reader
Brain/tissue atrophy quantification	High cross-sectionally; moderate longitudinally	Moderate in trials; protocol-sensitive	Sparse	Cleared volumetry tools exist	Absent; cost models assumption-heavy	Useful adjunct when acquisition and software versions are stable
Cord contour / spinal lesion analysis	High for cord; moderate for lesions	Moderate technical evidence	Absent	No MS-specific patient-benefit indication identified	Absent	Cord morphometry nearer use than lesion automation
Cortical lesions, CVS and PRL	Moderate in specialist data	Low to moderate; external sets small	Absent	No autonomous diagnostic indication identified	Absent	Research/specialist adjunct; end-to-end validation required
MS diagnosis / differential diagnosis	Moderate discrimination	Low; spectrum often artificial	Absent	No autonomous MS diagnosis clearance identified	Absent	Technical validation stage; image-only use is inappropriate
EDSS, PIRA, cognition or treatment prediction	Moderate and heterogeneous	Low; calibration rarely reported	Absent	No clinical prediction indication identified	Absent	Research stage; do not use for individual treatment choice

Note: Readiness integrates evidence quality rather than algorithm availability. Regulatory clearance is product-, version-, input- and intended-use-specific and is not equivalent to demonstrated workflow or patient benefit.

7. Research agenda

First, the field needs prospective, geographically distributed validation designed around intended use. Consecutive routine cases should be enrolled before outcomes are known, with patient-level and site-level separation, explicit missingness and adjudication of indeterminate scans. Brain and spinal cord protocols, contrast use and prior availability should reflect practice. Small-lesion and no-activity cases require deliberate sampling, and paediatric, progressive, comorbid and underrepresented populations should not be relegated to post hoc subgroup analysis.

Second, endpoints must move beyond aggregate overlap. For monitoring, prespecified outcomes could include actionable activity detected, false alerts per examination, reading time, confidence, report amendments, repeat imaging, treatment changes and inappropriate escalation. For prognosis, calibration and net benefit at defined time horizons are essential, as are clear time zero, treatment handling and stable definitions of progression. Patient outcomes may require cluster-randomised or stepped-wedge implementation studies, because the intervention is a human–AI workflow rather than a standalone model. The ongoing three-centre AssistMS randomised study is an important example because it includes disease-activity detection, treatment changes, reporting time and incremental cost per quality-adjusted life-year; its planned completion after the review cutoff means that no outcome claim can yet be made [83].

Third, open benchmarks should evolve from static leaderboards to versioned test ecosystems. Hidden multi-centre sets, longitudinal negative cases, scanner upgrades and adjudicated failure subsets would make overfitting harder. Data cards should report unique patients separately from sessions, demographics, disease phenotype, therapy, site, scanner, sequence parameters, preprocessing, rater process, licence and intended use. Evaluation servers can support restricted images, but results should remain auditable and repeated submissions controlled. Public challenges should include calibration, uncertainty and computational requirements as well as segmentation metrics.

Fourth, data collaboration should be paired with reproducibility. Federated learning, privacy-preserving analysis and common data models can access distributed cohorts, but should be compared with simpler site-diverse training and external evaluation. Where proprietary trials drive development, an independent public or regulator-accessible test should be prespecified. Models should be distributed with executable preprocessing and versioned weights whenever permissible. Negative external results and failure analyses deserve publication because they define the boundary of safe use.

Finally, imaging AI should be developed as measurement infrastructure rather than as a substitute diagnostician. Modular systems can expose image quality, registration, lesion candidates, volumes and uncertainty so that each component is testable. Diagnostic or prognostic layers should integrate clinical and laboratory information only when those inputs are available at the intended decision point. The most valuable model may not be the largest or most complex; it may be the one that is stable across time, abstains appropriately, reduces cognitive load and generates measurements clinicians and patients can understand.

8. Conclusion

AI in MS MRI has progressed from small algorithm demonstrations to multi-centre segmentation systems, longitudinal readers and quantitative pipelines with early workflow evidence. Brain white-matter lesion segmentation, tissue quantification and detection of new or enlarging lesions are closest to clinical translation when used as assistive tools with visible human verification. Cord contouring is robust, but spinal lesions, cortical lesions, enhancing lesions, CVS and PRL remain more dependent on specialised data, lesion size and acquisition quality. Diagnostic and prognostic models are scientifically informative but often overstate readiness through curated spectra, incomplete external validation, uncertain labels and absent calibration.

The decisive gap is no longer whether an algorithm can improve a benchmark metric. It is whether the entire data-to-decision pathway remains reliable across patients, scanners, time and clinical consequences. Regulatory clearance and faster reading are meaningful steps, but evidence of patient net benefit, cost-effectiveness and equitable performance is still sparse. Clinical translation should therefore proceed through prospective human–AI evaluation, transparent failure handling, independent multi-centre testing and continuous monitoring. On present evidence, AI is best positioned to make MS MRI measurement more consistent and longitudinal interpretation more sensitive—not to replace multidisciplinary diagnosis or independently determine prognosis.

Declarations

Funding: [Insert funding statement, including grant numbers, or “No specific funding was received for this work.”]

Conflict of interest: [Insert complete author disclosures.]

Author contributions: [Insert CRediT roles for all authors.]

Data availability: No new patient data were created or analysed for this review. Public sources and identifiers are provided in the references and dataset table.

Ethics approval and consent: Not applicable to this literature review.

AI assistance disclosure: This research draft was prepared with AI assistance for literature organisation, drafting and figure generation. Human authors must independently verify the search, evidence interpretation, numerical results, citations and compliance with the target journal’s current policies before any submission.

Acknowledgements: [Insert acknowledgements.]

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